

 INAF ISTITUTO NAZIONALE DI ASTRONOMIA OSSERVATORIO ASTRONOMICO DI ARCETRI	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 02-01-2025	 CONSORZIO LaMMA
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
December 2024



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1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2956	1154
	4.2 < ws < 7.6	1873	2212
	ws > 7.6	217	793
Number of points 13911 ; PC=64.3%; EBD=2.7%; POD1=58.6%; POD2=53.2%; POD3=80.3%			

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	4701	276
	4.2 < ws < 7.6	345	3668
	ws > 7.6	0	215
Number of points 13911 ; PC=92.2%; EBD=0.0%; POD1=93.2%; POD2=88.2%; POD3=94.7%			

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	12088	126
	12.0 < ws < 18.0	98	1265
	ws > 18.0	0	61
Number of points 13911 ; PC=97.7%; EBD=0.0%; POD1=99.2%; POD2=87.1%; POD3=85.7%			

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	6384	315	0		105
	East	429	678	143		0
	South	39	251	1536		111
	West	276	70	8		197
Number of points 10542 ; PC=83.4%; EBD=1.0%; POD1=89.6%; POD2=51.6%; POD3=91.0%; POD4=47.7%						

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	7021	78	0		29
	East	88	1195	40		0
	South	0	41	1639		11
	West	19	0	8		373
Number of points 10542 ; PC=97.0%; EBD=0.0%; POD1=98.5%; POD2=90.9%; POD3=97.2%; POD4=90.3%						

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3763	1253	9
	7.1 < rh < 15.3	2409	2961	488
	rh > 15.3	230	945	1569
Number of points 13627 ; PC=60.9%; EBD=1.8%; POD1=58.8%; POD2=57.4%; POD3=75.9%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	6073	271	0
	7.1 < rh < 15.3	329	4810	61
	rh > 15.3	0	78	2005
Number of points 13627 ; PC=94.6%; EBD=0.0%; POD1=94.9%; POD2=93.2%; POD3=97.0%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	13585	6	0
	50.0 < rh < 70.0	3	33	0
	rh > 70.0	0	0	0
Number of points 13627 ; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=84.6%; POD3=NaN%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

FATE - Monthly Report

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6227	1238
	1.9 < pwv < 3.5	923	2696
	pwv > 3.5	29	334
Number of points 13780 ; PC=78.6%; EBD=0.3%; POD1=86.7%; POD2=63.2%; POD3=81.7%			

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	7068	151
	1.9 < pwv < 3.5	111	4074
	pwv > 3.5	0	43
Number of points 13780 ; PC=97.5%; EBD=0.0%; POD1=98.5%; POD2=95.5%; POD3=98.5%			

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1398	1385	395
	0.6 < see < 0.8	917	1287	1037
	see > 0.8	99	363	859
Number of points 7740 ; PC=45.8%; EBD=6.4%; POD1=57.9%; POD2=42.4%; POD3=37.5%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2412	501	7
	0.6 < see < 0.8	345	2251	202
	see > 0.8	4	306	2094
Number of points 7740 ; PC=83.2%; EBD=0.1%; POD1=87.4%; POD2=73.6%; POD3=90.9%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2330	0	239
	0.6 < see < 0.8	0	2954	239
	see > 0.8	84	81	2052
Number of points 7740 ; PC=94.8%; EBD=4.2%; POD1=96.5%; POD2=97.3%; POD3=81.1%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2757	10	7
	0.6 < see < 0.8	0	3039	7
	see > 0.8	4	9	2296
Number of points 7740 ; PC=99.6%; EBD=0.1%; POD1=99.9%; POD2=99.4%; POD3=99.4%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1089	743	55
	$3.4 < \tau < 5.8$	431	1676	848
	$\tau > 5.8$	27	562	1303
Number of points 6734 ; PC=60.4%; EBD=1.2%; POD1=70.4%; POD2=56.2%; POD3=59.1%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1429	162	1
	$3.4 < \tau < 5.8$	118	2635	224
	$\tau > 5.8$	0	184	1981
Number of points 6734 ; PC=89.8%; EBD=0.0%; POD1=92.4%; POD2=88.4%; POD3=89.8%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1421	135	388
	$3.4 < \tau < 5.8$	99	2606	388
	$\tau > 5.8$	27	240	1818
Number of points 6734 ; PC=86.8%; EBD=6.2%; POD1=91.9%; POD2=87.4%; POD3=70.1%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1547	0	13
	$3.4 < \tau < 5.8$	0	2967	13
	$\tau > 5.8$	0	14	2193
Number of points 6734 ; PC=99.6%; EBD=0.2%; POD1=100.0%; POD2=99.5%; POD3=98.8%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1314	1419
	0.6 < glf < 0.7	605	833
	glf > 0.7	81	360
Number of points 6857 ; PC=37.9%; EBD=13.1%; POD1=65.7%; POD2=31.9%; POD3=20.0%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1680	392
	0.6 < glf < 0.7	307	1892
	glf > 0.7	13	328
Number of points 6857 ; PC=78.9%; EBD=0.3%; POD1=84.0%; POD2=72.4%; POD3=81.9%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1905	416
	0.6 < glf < 0.7	14	2178
	glf > 0.7	81	18
Number of points 6857 ; PC=79.5%; EBD=14.0%; POD1=95.2%; POD2=83.4%; POD3=46.4%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1978	26
	0.6 < glf < 0.7	9	2584
	glf > 0.7	13	2
Number of points 6857 ; PC=99.1%; EBD=0.3%; POD1=98.9%; POD2=98.9%; POD3=99.2%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2551	675
	4.6 < ws < 7.4	2049	2940
	ws > 7.4	176	1081
Number of points 14321 ; PC=65.6%; EBD=2.1%; POD1=53.4%; POD2=62.6%; POD3=80.5%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	4222	324
	4.6 < ws < 7.4	552	4123
	ws > 7.4	2	249
Number of points 14321 ; PC=90.6%; EBD=0.0%; POD1=88.4%; POD2=87.8%; POD3=95.4%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	11987	151
	12.0 < ws < 18.0	110	1519
	ws > 18.0	0	64
Number of points 14321 ; PC=97.3%; EBD=0.0%; POD1=99.1%; POD2=87.6%; POD3=86.3%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	5244	396	0		124
	East	83	338	46		0
	South	26	17	891		342
	West	1706	24	120		3082
Number of points 12439 ; PC=76.8%; EBD=0.4%; POD1=74.3%; POD2=43.6%; POD3=84.3%; POD4=86.9%						

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	6822	50	2		135
	East	54	718	8		0
	South	0	5	1004		32
	West	183	2	43		3381
Number of points 12439 ; PC=95.9%; EBD=0.0%; POD1=96.6%; POD2=92.6%; POD3=95.0%; POD4=95.3%						

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4552	1093	8
	8.9 < rh < 17.9	2336	3657	672
	rh > 17.9	84	530	1334
Number of points 14266 ; PC=66.9%; EBD=0.6%; POD1=65.3%; POD2=69.3%; POD3=66.2%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	6642	299	0
	8.9 < rh < 17.9	330	4891	78
	rh > 17.9	0	90	1936
Number of points 14266 ; PC=94.4%; EBD=0.0%; POD1=95.3%; POD2=92.6%; POD3=96.1%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	14186	10	0
	50.0 < rh < 70.0	7	59	4
	rh > 70.0	0	0	0
Number of points 14266 ; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=85.5%; POD3=0.0%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6527	930	0
	2.1 < pwv < 3.7	1218	2779	549
	pwv > 3.7	51	283	1890
Number of points 14227 ; PC=78.7%; EBD=0.4%; POD1=83.7%; POD2=69.6%; POD3=77.5%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	7641	75	0
	2.1 < pwv < 3.7	155	3866	45
	pwv > 3.7	0	51	2394
Number of points 14227 ; PC=97.7%; EBD=0.0%; POD1=98.0%; POD2=96.8%; POD3=98.2%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	3195	1261
	4.2 < ws < 7.6	1774	2126
	ws > 7.6	277	859
Number of points 14204 ; PC=63.8%; EBD=3.0%; POD1=60.9%; POD2=50.1%; POD3=79.5%			

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	6487	327	17
	East	373	704	173
	South	71	209	1532
	West	260	73	13

Number of points 10685 ; PC=83.8%; EBD=1.5%; POD1=90.2%; POD2=53.6%; POD3=88.3%; POD4=51.8%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3722	1245
	7.1 < rh < 15.3	2594	3032
	rh > 15.3	276	959

Number of points 13898 ; PC=59.9%; EBD=2.3%; POD1=56.5%; POD2=57.9%; POD3=75.6%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6260	1435
	1.9 < pwv < 3.5	1015	2598
	pwv > 3.5	68	347

Number of points 14058 ; PC=74.8%; EBD=0.8%; POD1=85.3%; POD2=59.3%; POD3=70.7%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1607	1574	517
	0.6 < see < 0.8	1141	1555	1228
	see > 0.8	169	551	1565
Number of points 9907 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2790	0	331
	0.6 < see < 0.8	0	3534	331
	see > 0.8	127	146	2979
Number of points 9907 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time ms		Observations		
		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1993	1059	147
	$3.4 < \tau < 5.8$	490	1915	923
	$\tau > 5.8$	42	673	1496
Number of points 8738 ; PC=61.8%; EBD=2.2%; POD1=78.9%; POD2=52.5%; POD3=58.3%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time ms		Observations		
		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2361	200	506
	$3.4 < \tau < 5.8$	122	3192	506
	$\tau > 5.8$	42	255	2060
Number of points 8738 ; PC=87.1%; EBD=6.3%; POD1=93.5%; POD2=87.5%; POD3=67.1%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1493	1667
	0.6 < glf < 0.7	1009	1403
	glf > 0.7	141	365
Number of points 8927 ; PC=39.9%; EBD=12.5%; POD1=56.5%; POD2=40.8%; POD3=23.4%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2470	546
	0.6 < glf < 0.7	32	2851
	glf > 0.7	141	38
Number of points 8927 ; PC=79.5%; EBD=13.6%; POD1=93.5%; POD2=83.0%; POD3=48.3%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2522	682	177
	4.6 < ws < 7.4	2083	2853	792
	ws > 7.4	188	1161	3880
Number of points 14338 ; PC=64.5%; EBD=2.5%; POD1=52.6%; POD2=60.8%; POD3=80.0%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	5256	463	13
	East	85	254	45
	South	36	41	826
	West	1682	17	172

Number of points 12439 ; PC=75.7%; EBD=0.5%; POD1=74.5%; POD2=32.8%; POD3=78.2%; POD4=86.8%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4217	1029
	8.9 < rh < 17.9	2599	3521
	rh > 17.9	156	730

Number of points 14266 ; PC=63.1%; EBD=1.3%; POD1=60.5%; POD2=66.7%; POD3=62.7%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6303	1069
	2.1 < pwv < 3.7	1412	2585
	pwv > 3.7	81	338

Number of points 14227 ; PC=74.4%; EBD=0.6%; POD1=80.8%; POD2=64.8%; POD3=69.7%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2650	1281	160
	4.2 < ws < 7.6	2171	1843	799
	ws > 7.6	337	1098	3731
Number of points 14070 ; PC=58.5%; EBD=3.5%; POD1=51.4%; POD2=43.7%; POD3=79.6%				

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	6294	329	17
	East	529	654	227
	South	159	268	1414
	West	209	49	34

Number of points 10630 ; PC=80.1%; EBD=2.1%; POD1=87.5%; POD2=50.3%; POD3=83.6%; POD4=34.9%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3445	1172
	7.1 < rh < 15.3	2741	3026
	rh > 15.3	320	990

Number of points 13764 ; PC=57.7%; EBD=3.4%; POD1=53.0%; POD2=58.3%; POD3=71.0%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6196	1551
	1.9 < pwv < 3.5	847	2316
	pwv > 3.5	143	485

Number of points 13873 ; PC=71.2%; EBD=1.5%; POD1=86.2%; POD2=53.2%; POD3=58.2%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1544	1320	462
	0.6 < see < 0.8	1111	1583	1147
	see > 0.8	275	731	1631
Number of points 9804 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2730	0	290
	0.6 < see < 0.8	0	3504	290
	see > 0.8	200	130	2950
Number of points 9804 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1925	1066	153
	$3.4 < \tau < 5.8$	555	1874	953
	$\tau > 5.8$	42	633	1445
Number of points 8646 ; PC=60.7%; EBD=2.3%; POD1=76.3%; POD2=52.4%; POD3=56.6%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2289	168	526
	$3.4 < \tau < 5.8$	191	3127	526
	$\tau > 5.8$	42	278	2025
Number of points 8646 ; PC=86.1%; EBD=6.6%; POD1=90.8%; POD2=87.5%; POD3=65.8%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1374	1480
	0.6 < glf < 0.7	1046	1527
	glf > 0.7	184	397
Number of points 8835 ; PC=41.2%; EBD=12.3%; POD1=52.8%; POD2=44.9%; POD3=26.2%			

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2379	452
	0.6 < glf < 0.7	41	2922
	glf > 0.7	184	30
Number of points 8835 ; PC=81.2%; EBD=12.9%; POD1=91.4%; POD2=85.8%; POD3=52.9%			

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2344	667	168
	4.6 < ws < 7.4	2213	2831	805
	ws > 7.4	236	1198	3876
Number of points 14338 ; PC=63.1%; EBD=2.8%; POD1=48.9%; POD2=60.3%; POD3=79.9%				

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

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wind direction degree	Observations			
	North	East	South	West
Model data	North	5114	462	20
	East	88	252	38
	South	90	34	810
	West	1767	27	188

Number of points 12439 ; PC=73.6%; EBD=1.1%; POD1=72.4%; POD2=32.5%; POD3=76.7%; POD4=83.8%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3932	935
	8.9 < rh < 17.9	2792	3447
	rh > 17.9	248	898

Number of points 14266 ; PC=60.7%; EBD=1.9%; POD1=56.4%; POD2=65.3%; POD3=63.7%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6095	1188
	2.1 < pwv < 3.7	1601	2345
	pwv > 3.7	100	459

Number of points 14227 ; PC=71.5%; EBD=0.9%; POD1=78.2%; POD2=58.7%; POD3=71.3%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration